Table 6: **(Q1)**: Metrics to assess the goodness of *layer 1* and *31* to detect personas (Level 2) in *Mistral-7B-Instruct*. Metrics: Silhouette Score (SH), Calinski-Harabasz Score (CH), Euclidean Distance (ED), Davies-Bouldin Score (DB).

Model	Dimension	layer	SH (↑)	CH (↑)	DB (↓)	ED (↑)
Mistral-7B-Instruct	AGREE	1	0.370 ± 0.082	176.5 ± 110.62	0.690 ± 0.084	0.110 ± 0.006
		31	0.251 ± 0.000	164.1 ± 0.00	1.743 ± 0.0008	5.403 ± 0.003
	CONSC	1	0.242 ± 0.113	96.8 ± 17.79	1.753 ± 0.505	0.063 ± 0.02
		31	0.171 ± 0.010	114.6 ± 9.23	2.140 ± 0.086	4.249 ± 0.122
	OPEN	1	0.423 ± 0.020	176.8 ± 13.84	1.006 ± 0.596	0.107 ± 0.023
		31	0.235 ± 0.000	194.2 ± 0.01	1.693 ± 0.000	5.147 ± 0.000
	EXTRA	1	0.459 ± 0.041	172.6 ± 39.78	0.558 ± 0.096	0.122 ± 0.020
		31	0.327 ± 0.000	225.2 ± 0.000	1.102 ± 0.000	8.020 ± 0.000
	NEURO	1	0.392 ± 0.109	123.9 ± 7.74	0.877 ± 0.573	0.107 ± 0.032
		31	0.203 ± 0.001	137.4 ± 3.31	1.873 ± 0.143	5.182 ± 0.271
	VIRTUE	1	0.389 ± 0.098	110.1 ± 7.74	0.961 ± 0.666	0.104 ± 0.032
		31	0.272 ± 0.000	211.5 ± 1.18	1.448 ± 0.056	5.499 ± 0.079
	RELAT	1	0.468 ± 0.077	378.1 ± 226.82	0.794 ± 0.070	0.119 ± 0.005
		31	0.226 ± 0.013	126.5 ± 13.76	1.370 ± 0.081	6.602 ± 0.295
	DEONT	1	0.392 ± 0.085	201.0 ± 120.04	0.740 ± 0.175	0.123 ± 0.011
		31	0.242 ± 0.001	155.4 ± 0.31	1.587 ± 0.082	5.631 ± 0.118
	UTILI	1	0.445 ± 0.064	188.0 ± 62.33	0.563 ± 0.087	0.135 ± 0.012
		31	0.319 ± 0.000	289.4 ± 0.000	1.215 ± 0.000	6.743 ± 0.000
	NIHIL	1	0.529 ± 0.000	436.5 ± 0.000	0.555 ± 0.000	0.130 ± 0.000
		31	0.177 ± 0.011	132.1 ± 17.07	2.058 ± 0.139	5.067 ± 0.278
	CONS	1	0.360 ± 0.098	98.9 ± 25.45	1.253 ± 0.559	0.083 ± 0.026
		31	0.230 ± 0.004	152.9 ± 0.31	1.843 ± 0.023	4.927 ± 0.072
	LIBER	1	0.511 ± 0.026	119.4 ± 49.65	0.495 ± 0.100	0.119 ± 0.002
		31	0.189 ± 0.026	119.4 ± 5.45	1.862 ± 0.326	4.769 ± 0.798
	LGBTQ	1	0.448 ± 0.041	136.8 ± 40.46	0.585 ± 0.101	0.122 ± 0.006
		31	0.255 ± 0.000	215.0 ± 0.01	1.478 ± 0.002	6.092 ± 0.006
	IMMI	1	0.437 ± 0.034	140.0 ± 93.28	0.558 ± 0.046	0.126 ± 0.009
		31	0.200 ± 0.006	149.7 ± 11.71	1.928 ± 0.080	4.945 ± 0.155

Table 7: **(Q1)**: Metrics to assess the goodness of *layer 1* and *31* to detect personas (Level 2) in *Granite-7B-Instruct*. Metrics: Silhouette Score (SH), Calinski-Harabasz Score (CH), Euclidean Distance (ED), Davies-Bouldin Score (DB).

Model	Dimension	layer	SH (↑)	CH (↑)	DB (↓)	ED (↑)
Granite-7B-Instruct	AGREE	1	0.420 ± 0.0312	210.7 ± 75.01	0.925 ± 0.225	0.287 ± 0.019
		31	0.611 ± 0.0000	969.0 ± 0.000	0.605 ± 0.000	17.62 ± 0.000
	CONSC	1	0.561 ± 0.0000	578.4 ± 0.003	0.709 ± 0.000	0.250 ± 0.000
		31	0.625 ± 0.0000	1033.5 ± 0.002	0.573 ± 0.000	16.83 ± 0.000
	OPEN	1	0.491 ± 0.0000	312.4 ± 0.000	0.912 ± 0.000	0.265 ± 0.000
		31	0.991 ± 0.0000	37132.7 ± 0.038	0.004 ± 0.000	2073.7 ± 0.000
	EXTRA	1	0.529 ± 0.0000	397.0 ± 0.000	0.820 ± 0.000	0.265 ± 0.000
		31	0.618 ± 0.0000	938.7 ± 0.000	0.595 ± 0.000	18.58 ± 0.000
	NEURO	1	0.491 ± 0.0362	270.3 ± 156.9	0.707 ± 0.157	0.279 ± 0.044
		31	0.637 ± 0.0000	1128.8 ± 0.006	0.549 ± 0.000	18.66 ± 0.000
	VIRTUE	1	0.533 ± 0.0001	350.0 ± 0.002	0.803 ± 0.000	0.242 ± 0.000
		31	0.996 ± 0.0000	164373.4 ± 0.028	0.000 ± 0.000	27948.3 ± 0.000
	RELAT	1	0.491 ± 0.0000	332.1 ± 0.000	0.927 ± 0.000	0.256 ± 0.000
		31	0.614 ± 0.000	828.7 ± 0.000	0.602 ± 0.000	17.32 ± 0.000
	DEONT	1	0.462 ± 0.0023	171.7 ± 65.54	0.678 ± 0.241	0.335 ± 0.048
		31	0.605 ± 0.0000	871.0 ± 0.005	0.622 ± 0.000	17.42 ± 0.000
	UTILI	1	0.481 ± 0.0000	286.0 ± 0.001	0.956 ± 0.000	0.262 ± 0.000
		31	0.994 ± 0.0000	80355.2 ± 0.057	0.002 ± 0.000	3690.19 ± 0.000
	NIHIL	1	0.441 ± 0.0208	184.3 ± 65.67	0.730 ± 0.229	0.300 ± 0.025
		31	0.602 ± 0.0000	292.7 ± 0.000	0.678 ± 0.000	18.44 ± 0.000
	CONS	1	0.511 ± 0.0659	284.2 ± 160.3	0.683 ± 0.126	0.276 ± 0.025
		31	0.993 ± 0.0000	8423.03 ± 0.000	0.001 ± 0.000	21282.9 ± 0.000
	LIBER	1	0.536 ± 0.0529	378.8 ± 236.3	0.624 ± 0.104	0.275 ± 0.039
		31	0.993 ± 0.0000	72452.4 ± 0.017	0.002 ± 0.000	3292.78 ± 0.000
	LGBTQ	1	0.497 ± 0.0509	290.2 ± 167.2	0.682 ± 0.126	0.283 ± 0.039
		31	0.593 ± 0.0000	838.5 ± 0.010	0.643 ± 0.000	15.95 ± 0.000
	IMMI	1	0.510 ± 0.0000	346.1 ± 0.000	0.871 ± 0.000	0.257 ± 0.000
		31	0.617 ± 0.0000	989.7 ± 0.000	0.594 ± 0.000	17.32 ± 0.000

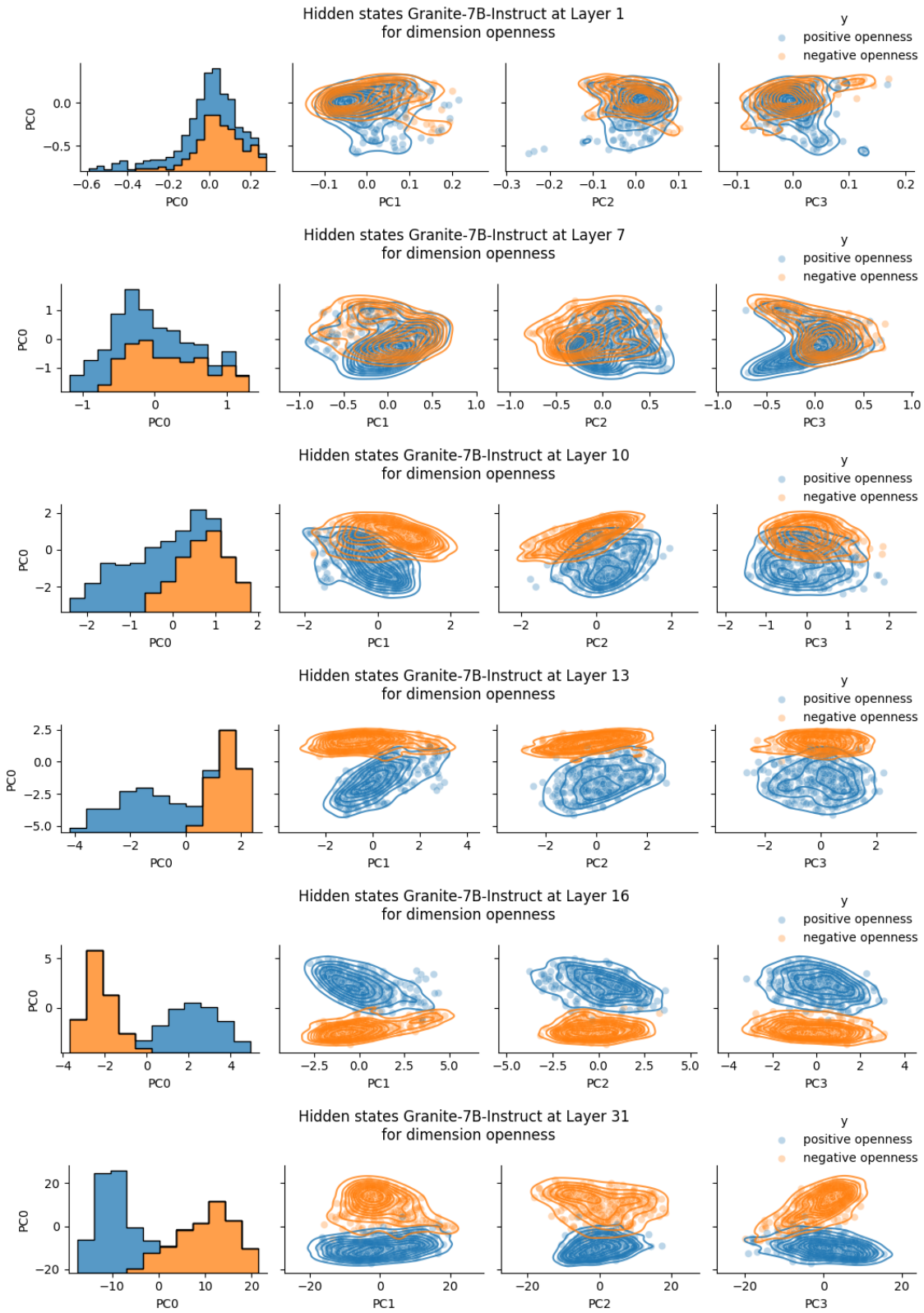

Figure 7: **(Q1)**: Examples of changes in the representation space of the dimension *openness* in a *Granite-7B-Instruct* model. Positive *conscientiousness* referring to MATCHINGBEHAVIOR, negative *conscientiousness* referring to NOTMATCHINGBEHAVIOR. We observe better separability in later layers.